

AI/ML Task 2 — Methodology & Results Report

Feature Engineering, Model Optimization & Performance Comparison | Maincrafts Technology

1. Objective

This report documents the end-to-end Machine Learning pipeline developed for Task 2 of the Maincrafts AI/ML internship program. The goal was to build an enhanced House Price Prediction system using feature engineering, preprocessing, multi-model training, and structured performance comparison.

2. Dataset

The California Housing Dataset was used, consisting of 20,640 samples with 8 input features and Median House Value as the prediction target.

Feature	Description
MedInc	Median income of block group
HouseAge	Median house age in block group
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average household occupancy
Latitude	Block group latitude
Longitude	Block group longitude

3. Methodology

3.1 Feature Scaling

StandardScaler was applied to normalize all input features to a mean of 0 and standard deviation of 1. This is a critical preprocessing step — without it, high-range features like Population would dominate gradient-based learning, leading to unstable and biased model performance.

3.2 Train-Test Split

The dataset was split into 80% training data (16,512 samples) and 20% test data (4,128 samples) using random_state=42 for reproducibility. The test set ensures evaluation on unseen data, preventing data leakage.

3.3 Models Trained

- Linear Regression** — Baseline model assuming a linear relationship between features and target.
- Ridge Regression (alpha=1.0)** — Adds L2 regularization to penalize large coefficients and reduce overfitting.
- Decision Tree Regressor (max_depth=5)** — Captures non-linear relationships by recursively splitting feature space.

3.4 Evaluation Metrics

Two metrics were used: RMSE (Root Mean Squared Error) penalizes large prediction errors and is in the same unit as the target variable; lower values indicate better accuracy. R^2 Score (coefficient of determination) measures the proportion of variance explained by the model; values closer to 1.0 are better.

4. Results

The table below summarises model performance on the held-out test set:

Model	RMSE	R^2 Score	Rank
Linear Regression	0.3944	0.9048	3rd
Ridge Regression	0.3944	0.9048	2nd
Decision Tree	0.3348	0.9313	1st (Best)

5. Conclusion & Model Selection

The Decision Tree Regressor (max_depth=5) was selected as the best-performing model based on its superior RMSE (0.3348) and R^2 Score (0.9313). Its ability to capture non-linear patterns in the housing data made it significantly more accurate than the linear models.

Linear Regression and Ridge Regression produced identical results in this experiment, suggesting that regularization had minimal impact, likely because multicollinearity was not severe in this dataset. In more complex real-world datasets, Ridge would typically outperform plain Linear Regression.

The best model was saved using joblib for future deployment or inference. This pipeline is reproducible, well-documented, and suitable for inclusion in a professional ML portfolio.